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To

Hiring Manager,

Fervo Energy,

910 Louisiana St., Suite #4400, Houston, TX, 77002

September 16, 2025

Sub: Application for Reservoir Engineering Internship

Dear Hiring Manager,

I am writing to express my enthusiastic interest in the Geophysics Intern position at Fervo Energy, as advertised. As a third-year Geophysics Ph.D. student at the Colorado School of Mines and a current collaborator on a EGS project with Fervo, I have developed a deep appreciation for your company's innovative work and believe my skills are exceptionally well-aligned with the requirements of this role.

My Ph.D. research on using Distributed Fiber-Optic Sensing (DFOS) for hydraulic fracture characterization directly matches this role's responsibilities. On an ExxonMobil-sponsored project, I developed a complete LF-DAS data processing workflow and built a coupled pressure-strain numerical model using MOOSE to perform history matching against field data. This experience solidified my ability to analyze large datasets and model subsurface fluid dynamics.

To support my research, I developed fiberRIS, an open-source Python package designed for the history matching and interpretation of DAS data, hence to characterize subsurface reservoirs. This project demonstrates my advanced proficiency in Python and my passion for creating practical tools for data analysis. Furthermore, my academic coursework in Advanced Machine Learning and Advanced Data Science has provided me with a strong foundation to explore the scientific machine learning approaches mentioned in the job description.

Living locally, I am excited by the prospect of contributing to your Golden team. The opportunity to apply my expertise in DAS analysis, numerical modeling, and Python development to advance Fervo Energy's geothermal projects is incredibly compelling.

Thank you for your consideration. I have attached my resume and welcome the opportunity to discuss how my experience can benefit Fervo Energy.

Sincerely,

Shenyao Jin